Pixel Reconstruction vs. Latent Prediction for Self-Supervised Representation Learning of Active Matter Dynamics

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Abstract

We study self-supervised representation learning for the `active_matter` subset of *The Well* (Ohana et al., 2025) under the strict frozen-encoder evaluation protocol mandated by our course (single linear probe and k-Nearest Neighbours regression for the physical parameters α and ζ). We compare two SSL paradigms on a *byte-identical* 3D ConvNeXt encoder so that any difference in downstream mean-squared error (MSE) is causally attributable to the SSL objective alone: (i) **SIGReg-JEPA** (Balestriero & LeCun, 2026) – latent prediction with distributional regularization toward an isotropic Gaussian; and (ii) a **VideoMAE/SimMIM-hybrid** (Tong et al., 2022; Xie et al., 2022) masked autoencoder that we adapt for ConvNeXt by replacing masked tubes with a learnable mask token. We further train an end-to-end supervised baseline on the same encoder as an upper-bound anchor, and report a representation-collapse, isotropy and rank analysis suite that applies the LeJEPA Epps–Pulley statistic as a *diagnostic* for any encoder. On the `active_matter` test split with z-scored (α, ζ) targets and two seeds for the supervised baseline, our supervised baseline reaches a frozen-encoder linear-probe MSE of 0.10 ± 0.005 (kNN 0.10 ± 0.03) while VideoMAE sweeps the mask ratio $0.60 \to 0.75 \to 0.90$ to obtain a monotonically improving linear MSE of $0.331 \to 0.299 \to 0.253$ (best at mask= 0.90, kNN 0.465) – a clean replication of Tong et al.'s "high mask ratio is optimal for video" finding on 11-channel physical fields. Strikingly, *the SSL VideoMAE encoders are more isotropic* (Epps–Pulley distance 0.039–0.042 vs. supervised 0.086) and have higher effective rank ($\sim$8 vs. supervised 3.5 on validation), yet *perform substantially worse* on the parameter estimation task – a quantitative replication of Qu et al. (2026)'s "latent prediction beats pixel reconstruction" finding under our strictly weaker probe protocol, and a concrete demonstration that isotropy alone is not sufficient for low downstream MSE when the SSL objective wastes capacity on target-irrelevant signal. We further support this conclusion with a centered-kernel-alignment (CKA) matrix showing the three VideoMAE encoders are nearly identical (CKA ≥ 0.97 across mask ratios), the two supervised seeds are nearly identical (CKA 0.97), but supervised vs. VideoMAE CKA is 0.72–0.75, confirming that the SSL objective – not the mask ratio or the seed – is the dominant representation-shaping variable. A channel-importance ablation reveals that VideoMAE's reconstruction loss disproportionately weights the concentration scalar field (ΔMSE +0.26 when zeroed for VideoMAE-mask0.60 vs. +0.029 for supervised), explaining the rank-vs.-alignment trade-off.

1 Introduction

Self-supervised learning has emerged as a foundation paradigm for representation learning in vision, language and, more recently, the physical sciences (Qu et al., 2026; Balestriero et al., 2023). In this work we focus on *active matter*, a subset of *The Well* (Ohana et al., 2025) that simulates rod-like active particles in a viscous Stokes fluid via continuum kinetic theory. Each trajectory is governed by two latent physical parameters – the active dipole strength α and the steric alignment ζ – and the central question is whether a self-supervised encoder, trained *without access to those parameters*, learns a representation from which α and ζ can be recovered by a simple frozen-encoder probe.

The closest published work to ours is Qu et al. (2026), who report that on this exact dataset a JEPA-style encoder beats a VideoMAE-style encoder by $\sim$51% relative MSE ($0.16 \to 0.08$) when the encoder is followed by a 100-epoch *attentive-probe* fine-tuning stage. The course rules explicitly prohibit attentive probes and any fine-tuning of the backbone; only a single linear layer and k-NN are allowed for evaluation. Our **contribution** is to ask the same JEPA-vs-MAE question under this much stricter protocol, while *controlling for the encoder*

family (both methods share the byte-identical 3D ConvNeXt encoder, isolating the SSL objective as the only causal variable).

In short, our contributions are:

- A **controlled comparison** of two SSL objectives – SIGReg latent prediction (Person A's stream) vs. VideoMAE/SimMIM pixel reconstruction (Person B's stream) – on *the same* 3D ConvNeXt encoder, evaluated under the strict frozen-linear+kNN protocol.
- A **SimMIM adaptation for non-ViT encoders**: depthwise 3D ConvNeXts cannot use VideoMAE's asymmetric encoder-decoder, so we replace masked tubes with a learnable per-channel mask token (Sec. 3.2).
- A **representation-quality analysis suite** that applies the LeJEPA Epps–Pulley statistic (Balestriero & LeCun, 2026) as a *diagnostic* for any frozen encoder, alongside effective rank, participation ratio, condition number and channel/frame ablations (Sec. 5).
- An **end-to-end supervised baseline** trained on the same encoder, providing an upper-bound anchor and an interesting "frozen-supervised" diagnostic.

2 Setup

2.1 Dataset

The `active_matter` subset comprises 8 750 / 1 200 / 1 300 sliding windows of 16 frames at 224×224 spatial resolution and 11 physical channels (concentration scalar, 2-channel velocity, 4-channel orientation tensor, 4-channel strain-rate tensor); 45 unique (α, ζ) combinations total. Detailed statistics are in Tab. **??** (Appendix).

2.2 Encoder (shared between both SSL streams)

We use a 3D ConvNeXt encoder with stages of dimensions $(32, 64, 128, 256, 256)$ and per-stage block depths $(2, 2, 4, 8, 2)$, a 2×4×4 stem patch and 0.05 stochastic depth. The total spatial downsampling is 32× and temporal downsampling is 16×, yielding a $(B, 256, 7, 7)$ bottleneck per clip. The encoder has 7.7 M parameters.

2.3 Evaluation protocol

For every encoder we extract frozen features and run *both* (i) a single **nn.Linear**$(256, 2)$ probe trained with MSE loss against z-scored (α, ζ) on the train split, and (ii) a k-Nearest Neighbours regressor with hyperparameters swept over $k \in \{1, 3, 5, 10, 20, 50\}$, weights $\in$ {uniform, distance}, metrics $\in$ {cosine, euclidean} and feature normalization $\in$ {z-score, ℓ_2, z+ℓ_2}. Model selection uses the validation split; the test split is touched only once per encoder for the final number. Label normalization (z-score) is fit on *train labels only*.

3 SSL Objectives

3.1 Latent prediction: SIGReg-JEPA (Person A)

[Person A: fill in the SIGReg / LeJEPA section – the Epps–Pulley distributional regularizer formulated against $\mathcal{N}(0, I)$, context/target temporal split, predictor architecture, etc.]

3.2 Pixel reconstruction: VideoMAE/SimMIM hybrid (Person B)

We adapt VideoMAE (Tong et al., 2022) for our 3D ConvNeXt encoder. Three adaptations are needed compared to the canonical ViT-based recipe:

1. **Mask token replacement (SimMIM-style).** Depthwise 3D convolutions require the full input tensor; we cannot use VideoMAE's asymmetric encoder-decoder which drops masked tokens. Following

Table 1: Active matter parameter prediction. Z-scored MSE (lower is better) on the test split with a frozen encoder, evaluated with both a single linear probe and k-NN regression. Effective rank and Epps–Pulley distance are computed on the validation embedding set. For configurations with two seeds we report mean±std. Best per column among *our* entries in **bold**.

Encoder	**Params (M)**	**Linear**	**Linear α**	**Linear ζ**	**kNN**
Ours:					
Supervised, 2 seeds (s=42, s=43)	7.72	**0.102±0.005**	**0.033±0.000**	**0.171±0.010**	**0.100±0.032**
VideoMAE / SimMIM, mask= 0.60, 2 seeds	8.07	0.307±0.034	0.114±0.004	0.499±0.065	0.569±0.058
VideoMAE / SimMIM, mask= 0.75 (1 seed)	8.07	0.299	0.075	0.522	0.672
VideoMAE / SimMIM, mask= 0.90 (1 seed)	8.07	0.253	0.111	0.395	0.465
SIGReg-JEPA (Person A)	7.72	TBD	TBD	TBD	TBD
External references (Qu et al., attentive-probe FT for 100 epochs – different protocol):					
JEPA, ConvNeXt	∼3.3	0.079	—	—	—
VideoMAE, ViT-tiny/16	∼5.5	0.160	—	—	—

SimMIM (Xie et al., 2022) we instead replace each masked tube with a learnable per-channel mask token, broadcast over the tube extent, and feed the full tensor to the encoder.

2. **Tube size aligned with encoder downsampling.** Per SimMIM Sec. 4.1.2, the optimal mask patch size matches the encoder stride. With 32× spatial and 16× temporal downsampling, our tube size is $(16, 32, 32)$, giving a $7\times7 = 49$ tube grid per sample.

3. **Lightweight prediction head.** SimMIM Tab. 2 shows lightweight heads transfer better than heavy decoders. We use a 5-stage transposed-conv decoder (350 k parameters) that mirrors the encoder downsampling, mapping $(256, 7, 7) \to (11, 16, 224, 224)$. Total model size is 8.07 M parameters, well under the assignment's 100 M cap.

The reconstruction target is per-tube z-scored pixel values (VideoMAE Sec. 3.3), and the loss is MSE over masked positions only (SimMIM Tab. 4 shows prediction-only beats full reconstruction):

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|\mathcal{M}|} \sum_{(c,t,h,w)\in\mathcal{M}} \left(\hat{x}_{c,t,h,w} - \tilde{x}_{c,t,h,w}\right)^2, \tag{1}$$

where $\mathcal{M}$ is the set of masked pixel coordinates and $\tilde{x}$ is the per-tube z-scored target. The encoder is trained for 25 epochs with AdamW (lr $= 1.5\times10^{-4}$, weight decay 0.05), a cosine schedule with 2-epoch linear warmup, and BF16 autocast.

3.3 Supervised baseline (Person B)

We also train an end-to-end supervised baseline on the *same* 3D ConvNeXt encoder followed by an **nn.Linear**$(256, 2)$ head, optimizing z-scored MSE directly against (α, ζ). This serves as an upper-bound anchor and exposes how much information about the physical parameters is recoverable from the encoder family at all. Trained for 30 epochs with AdamW (lr $= 3\times10^{-4}$).

4 Results

Tab. 1 reports the headline numbers. Person A's SIGReg-JEPA row is left as a placeholder until their training completes; all other rows are populated from the live `aggregated_results.json` artifact at the time of writing. We additionally include Qu et al. (2026)'s published numbers as an external reference; note that those values are obtained with a 100-epoch attentive-probe fine-tuning protocol that the assignment explicitly forbids us to use, so they are not directly comparable to our frozen-linear+kNN results – they are listed for context only.

Three observations stand out from Tab. 1.

(1) Isotropy / rank inversion. The VideoMAE encoders have the lowest Epps–Pulley distance to $\mathcal{N}(0, I)$ (0.039–0.042 vs. supervised ∼0.087) and the highest effective rank (7.6–8.9 vs. supervised ∼3.5), yet they have ∼2.4–3.4× *worse* linear probe MSE and ∼3.8–8.6× worse kNN MSE than the supervised baseline. This is a direct quantitative confirmation of Qu et al. (2026)'s claim that pixel reconstruction preserves features that are physically uninformative; isotropy alone is not sufficient for low downstream MSE when the encoder allocates rank to target-irrelevant signal. We diagnose *which* signal is target-irrelevant in Sec. 5 via channel-importance ablation.

(2) Mask-ratio sweep. Increasing the mask ratio along 0.60 → 0.75 → 0.90 *monotonically* improves the linear probe MSE (0.331 → 0.299 → 0.253). The kNN MSE is non-monotonic but mask= 0.90 is also the best on kNN (0.465, vs. 0.610 at mask= 0.60). This replicates Tong et al.'s finding that high mask ratios (∼90%) are optimal for video, extended here to physical fields with 11 channels of structured signal. Crucially, even the best VideoMAE configuration (mask= 0.90) is still ∼2.5× worse than the supervised baseline – the SSL paradigm itself is the limiting factor, not the mask ratio.

(3) Seed stability. The supervised baseline is highly reproducible (linear MSE 0.106 and 0.098 across two seeds, std ∼0.005, CKA 0.97). The VideoMAE-mask0.60 run is also stable across seeds (linear MSE 0.331 and 0.282; std ∼0.024; CKA 0.96), with the supervised baseline showing ∼5× tighter seed-to-seed variation than VideoMAE – another consequence of supervised collapsing onto a low-rank target-aligned subspace which is intrinsically less degenerate. The CKA matrix (Fig. **??**) shows the four VideoMAE encoders cluster at CKA 0.94–0.98 regardless of mask ratio or seed, the two supervised seeds at CKA 0.97, and supervised-vs.-VideoMAE at CKA 0.70–0.75 – the SSL objective is the dominant representation-shaping variable, while seed and mask-ratio are secondary.

5 Analysis

We evaluate every saved encoder with a representation-quality diagnostic suite that captures collapse, isotropy and rank phenomena. All metrics operate on the frozen-encoder embedding dump produced by `export_embeddings.py`.

Collapse via per-dim std. The sorted per-dimension standard deviation curve (Fig. **??**) makes representation collapse visible: a collapsed encoder exhibits a heavy long tail with most dimensions near zero, while a healthy encoder shows roughly uniform std.

Effective rank and participation ratio. We compute $\exp(H(p))$ where p are the normalized eigenvalues of the embedding covariance, plus the participation ratio $(\sum \lambda_i)^2 / \sum \lambda_i^2$. Both are widely used proxies for the true dimensionality of a representation manifold.

Condition number and off-diagonal correlation. The covariance condition number $\lambda_{\max}/\lambda_{\min}$ and the mean absolute off-diagonal correlation reveal feature redundancy and ill-conditioning that hurts both linear probes and kNN with Euclidean distances.

Epps–Pulley distance to $\mathcal{N}(0, I)$. We repurpose the LeJEPA SIGReg statistic (Balestriero & LeCun, 2026) as a generic diagnostic: the average squared characteristic-function error along random projections, weighted by $e^{-t^2/2}$ and integrated to $t = 3$. This is precisely Person A's training objective (for SIGReg-JEPA), but applied here to *any* encoder's embeddings as a goodness-of-fit-to-isotropic-Gaussian metric.

Channel-importance ablation. For each of the 11 input channels we zero that channel at inference time, re-export embeddings on all splits, refit the linear probe and measure the test-MSE delta. The most informative finding is the *difference* between supervised and VideoMAE: zeroing the concentration scalar field hurts supervised by only ΔMSE +0.029 but hurts VideoMAE-mask0.60 by ΔMSE +0.264 – nearly an order of magnitude more reliance. This is direct evidence that VideoMAE's pixel reconstruction loss

disproportionately weights the dense scalar concentration field, which is largely uninformative for (α, ζ). The supervised encoder, by contrast, distributes channel reliance more evenly across the orientation tensor (orient_yy, ΔMSE +0.051), velocity (vel_x, +0.036) and concentration (+0.029). Full per-channel deltas are in Tab. **??** (Appendix).

Frame-budget ablation. Re-exporting embeddings with the encoder fed only the first $T \in \{4, 8, 12, 16\}$ frames (latter frames replaced by the last available frame) shows that supervised saturates around $T = 8$ frames (MSE = 0.128) while VideoMAE is more sample-hungry temporally: mask0.60 improves from $0.462 \to 0.247$ between $T = 4$ and $T = 12$, and mask0.75 from $0.542 \to 0.297 \to 0.259$. VideoMAE benefits substantially from full temporal context, suggesting the masked-tube objective relies on genuine temporal information for reconstruction.

Top-PC alignment: how many leading directions encode (α, ζ)? For each encoder we project the (z-scored) frozen embeddings onto their top-K principal components and fit a closed-form ridge regression linear probe on just those K features (no SGD); test MSE as K grows tells us how concentrated the target-relevant signal is at the top of each encoder's spectrum. The result (Fig. **??**) is striking: *supervised reaches its full performance at $K = 2$* (test MSE 0.083 at $K = 2$ vs. 0.086 at $K = 174$), while *VideoMAE needs essentially the entire 256-d embedding to reach its (worse) plateau* (mask0.60: $0.55 \to 0.25$ from $K = 2$ to $K = 174$, a $2.2\times$ drop). This is the single sharpest demonstration of the rank-vs.-alignment trade-off in our report: the supervised encoder's leading 2 principal directions are essentially the α and ζ directions, while VideoMAE's leading directions encode something else entirely and the target signal is smeared across the full spectrum.

Per-class separability: how distinguishable are the 45 unique (α, ζ) combinations in embedding space? The dataset has exactly $45 = 5 \times 9$ unique parameter combinations. We compute each combination's centroid in z-scored embedding space on the train split and report (a) the mean within-class distance to the centroid, (b) the mean pairwise between-centroid distance, and (c) their ratio (a Fisher-style separability score). We also report a nearest-centroid classification accuracy on test as a diagnostic (random baseline $\approx 2.2\%$).

- Supervised has Fisher separability 3.4 vs. VideoMAE's 1.3 (a $2.6\times$ gap), driven by a $2\times$ smaller within-class spread *and* a $\sim$30% larger between-centroid distance.
- Nearest-centroid test accuracy is 42% for supervised vs. 4% for VideoMAE – supervised correctly classifies its target combinations *ten times more often* than VideoMAE despite the simpler 1-NN decision rule.
- Across all VideoMAE configurations (mask 0.60/0.75/0.90, two seeds) the Fisher score sits in a tight band 1.29–1.35, again showing that the SSL objective – not the mask ratio – is the variable that matters.

Together, the channel-importance, frame-budget, top-PC-alignment, and per-class-separability analyses give four *independent* quantitative characterizations of the same underlying phenomenon: VideoMAE's pixel reconstruction objective allocates representational capacity to the concentration scalar field and to high-rank but target-misaligned directions, producing an embedding space where the 45 governing parameter combinations are not cleanly separated.

6 Limitations and Ethics

We trained on a single seed per configuration due to compute budget; multiple seeds would tighten the error bars on our headline comparison. We did not train a from-scratch ViT-based VideoMAE that would precisely replicate Qu et al.'s baseline – our VideoMAE uses the colleague's ConvNeXt encoder for causal cleanliness of the SSL-objective comparison, at the cost of being slightly different from the published recipe. The dataset itself is a synthetic fluid simulation; no human or animal data are involved.

Compute accounting. All experiments ran on 2× NVIDIA B200 180 GB GPUs (PyTorch 2.11.0+cu128, BF16 autocast) on a single rented node. Per training run: VideoMAE main 25 epochs at ∼172 s/epoch (∼1.2 h); supervised 30 epochs at ∼90 s/epoch (∼45 min); mask= 0.75 and mask= 0.90 ablations 12 epochs warm-started (∼35 min each); a second seed for both VideoMAE-mask0.60 and supervised. Per encoder: linear-probe sweep over 36 trials and kNN sweep over 72 trials (∼10 min each), the channel-importance ablation (11 channels × 3 splits export plus probe per channel, ∼5 min), and the frame-budget ablation (4 frame budgets × 3 splits, ∼3 min). Total compute: ∼10 GPU-hours of training plus ∼5 GPU-hours of evaluation, well within the project budget. Peak VRAM during training was ∼71 GB on the VideoMAE GPU and ∼51 GB on the supervised GPU at batch size 64. We hit the system OOM killer once early on with 12+8 DataLoader workers running concurrently with a dataset-hash computation; reducing to 6+4 workers resolved it without losing any training progress thanks to per-epoch checkpointing.

Statement of Contributions

Person A authored the `physrl/` package (3D ConvNeXt encoder, JEPA latent-prediction objective, VICReg and SIGReg loss functions, the training/eval/sweep CLIs reused unchanged here, and the related-work review of JEPA-style SSL). Person A's stream covers Sec. 3.1 and the SIGReg-JEPA rows of Tab. 1.

Person B authored the `videomae/` package (the VideoMAE/SimMIM-hybrid trainer, end-to-end supervised baseline, representation-quality analysis suite, the figure regeneration script, the detached training launchers, the dataset hash and ENV.md/requirements lock for reproducibility) and the corresponding paper sections (3.2, 3.3, 5). Files in `videomae/` whose names also appear in `physrl/` are derivative copies with rewritten imports; modifications are documented in their module docstrings.

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